

MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL-1902



DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No : 07
Lab Report Name : VLAN Configuration Using Two switch in cisco packet Tracer
Course Title : Computer Networks Lab
Course Code : ICT-3202

Submitted By	Submitted To
Name: Md. Mohtasim Billah Zim ID: IT23016 3rd Year 2nd Semester Session: 2022-2023 Department of ICT, MBSTU	Dr. Md. Abir Hossain Professor Department of ICT, MBSTU

Date of Performance:

Date of Submission:

Experiment No: 07

Experiment Name: VLAN Configuration Using Two switch in cisco packet tracer

Introduction:

Virtual Local Area Networks (VLANs) play a pivotal role in modern enterprise network infrastructure by logically partitioning a single physical Local Area Network (LAN) into multiple distinct broadcast domains. This logical segmentation enhances security, minimizes unnecessary broadcast traffic, and streamlines administrative controls without requiring physical recabling.

When a network spans multiple Ethernet switches, traffic originating from different VLANs must traverse shared inter-switch links. To achieve this, trunk links utilizing the IEEE 802.1Q encapsulation protocol are configured on switch-to-switch connections. An 802.1Q trunk tag inserts a 4-byte header into Ethernet frames to preserve their VLAN identity across switches. This lab experiment demonstrates the complete setup, configuration, CLI validation, and connectivity verification of two VLANs (VLAN 10 and VLAN 20) deployed across two Cisco Catalyst 2960 switches connected via an IEEE 802.1Q trunk link on FastEthernet0/1.

Objectives:

1. Design and simulate a dual-switch topology in Cisco Packet Tracer.
2. Configure dynamic/static access ports on switches and map them to dedicated VLANs (VLAN 10 and VLAN 20).
3. Establish an IEEE 802.1Q trunk link on port FastEthernet0/1 inter-connecting Switch0 and Switch1.
4. Assign static IPv4 addresses to host devices corresponding to their assigned subnets (192.168.10.0/24 and 192.168.20.0/24).
5. Verify intra-VLAN connectivity across switches and confirm Layer 2 isolation between different VLANs without routing.

Necessary Materials and Software:

The hardware and simulation tools utilized for this lab experiment are summarized below:

Equipment / Tool	Quantity	Description / Details
Cisco Packet Tracer	v8.x+	Network simulation platform
Cisco Catalyst 2960 Switch	2	24-Port FastEthernet Layer 2 Switches (Switch0, Switch1)
Generic End Devices (PCs)	4	Workstation hosts (PC0, PC1, PC2, PC3)
Copper Straight-Through Cables	4	Used for connecting end devices to switch access ports
Copper Crossover Cable	1	Used for inter-switch trunk link connection (Fa0/1 to Fa0/1)

Network Addressing & Port Mapping:

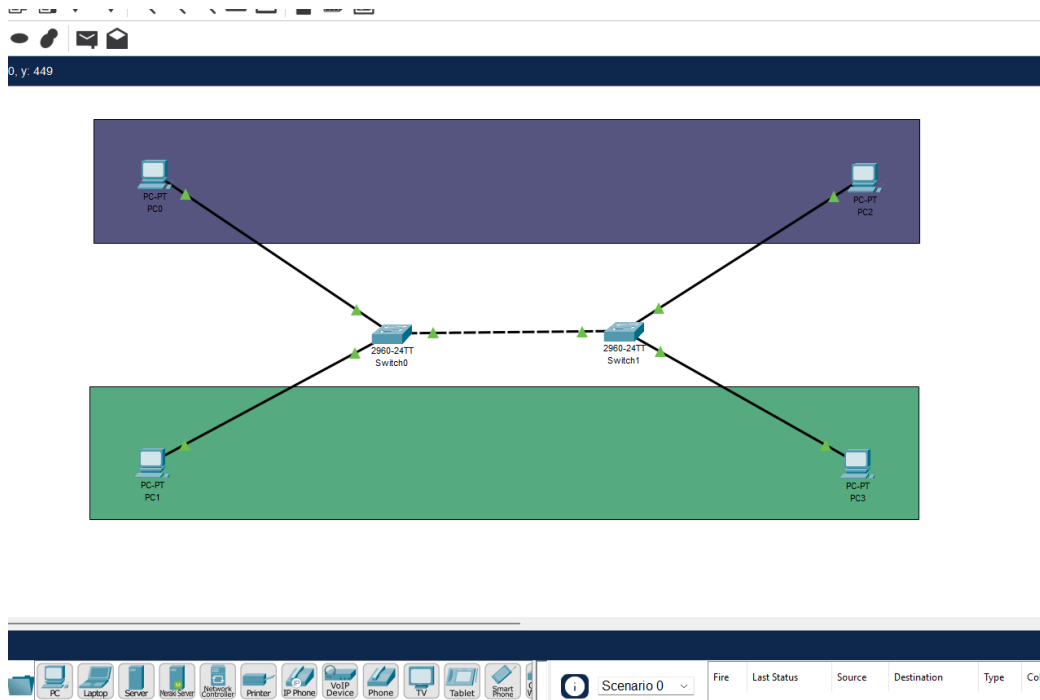
The detailed IPv4 network addressing scheme, VLAN assignments, and switch interface mappings are structured as follows:

Device	Interface	IP Address	Subnet Mask	VLAN Assignment	Switch Connection
PC0	FastEthernet0	192.168.10.10	255.255.255.0	VLAN 10 (VLAN0010)	Switch0 - Fa0/3
PC1	FastEthernet0	192.168.20.10	255.255.255.0	VLAN 20 (VLAN0020)	Switch0 - Fa0/2
PC2	FastEthernet0	192.168.10.20	255.255.255.0	VLAN 10 (VLAN0010)	Switch1 - Fa0/3
PC3	FastEthernet0	192.168.20.20	255.255.255.0	VLAN 20 (VLAN0020)	Switch1 - Fa0/2

Experimental Steps & Implementation Details:

Step 1: Network Topology Setup & Interconnection

In Cisco Packet Tracer, two 2960-24TT switches were placed in the workspace. Four PCs were deployed and connected to the switches. Switch0 and Switch1 were interconnected using a Copper Crossover Cable attached to FastEthernet0/1 on both ends. PC0 and PC1 were linked to Switch0 via FastEthernet0/3 and FastEthernet0/2, while PC2 and PC3 were connected to Switch1 via FastEthernet0/3 and FastEthernet0/2.



Step 2: Static Host IP Addressing Configuration

Each end device was configured statically through the Desktop > IP Configuration interface in Packet Tracer:

- **PC0 Configuration:** IPv4 Address: 192.168.10.10 | Subnet Mask: 255.255.255.0
- **PC1 Configuration:** IPv4 Address: 192.168.20.10 | Subnet Mask: 255.255.255.0
- **PC2 Configuration:** IPv4 Address: 192.168.10.20 | Subnet Mask: 255.255.255.0
- **PC3 Configuration:** IPv4 Address: 192.168.20.20 | Subnet Mask: 255.255.255.0

The screenshot shows the configuration window for PC0 in Packet Tracer. The 'Desktop' tab is selected, and the 'IP Configuration' interface is open. The 'Interface' dropdown is set to 'FastEthernet0'. The 'IP Configuration' section has 'Static' selected, with the following values: IPv4 Address: 192.168.10.10, Subnet Mask: 255.255.255.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. The 'IPv6 Configuration' section has 'Automatic' selected, with the following values: IPv6 Address: (empty), Link Local Address: FE80::202:16FF:FE1B:E025, Default Gateway: (empty), and DNS Server: (empty). The '802.1X' section has 'Use 802.1X Security' unchecked, with Authentication set to MD5, Username: (empty), and Password: (empty).

PC2

Physical Config Desktop Programming Attributes

P Configuration [X]

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static This address is already used in the network.

IPv4 Address 192.168.10.20

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20A:41FF:FE48:B014

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MDS

Username

Password

PC1

Physical Config Desktop Programming Attributes

P Configuration [X]

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.10

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20C:85FF:FEB7:B0AA

Default Gateway

DNS Server

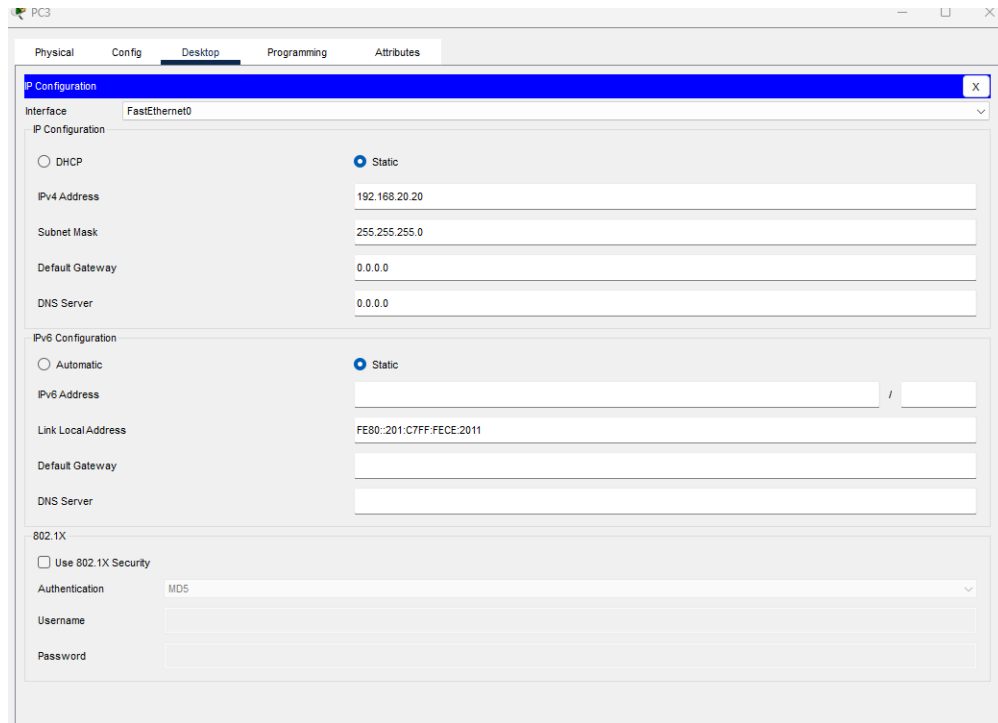
802.1X

☐ Use 802.1X Security

Authentication MDS

Username

Password



Step 3: VLAN Creation & Port Assignment CLI Commands

To isolate broadcast domains, VLAN 10 and VLAN 20 were created on both switches. Access ports were explicitly configured to assign individual switch interfaces to their respective VLANs:

--- CLI Configuration on Switch0 & Switch1 ---

```
Switch# configure terminal
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name VLAN0010
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 20
```

```
Switch(config-vlan)# name VLAN0020
```

```
Switch(config-vlan)# exit
```

! Assign FastEthernet0/3 to VLAN 10

```
Switch(config)# interface FastEthernet0/3
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 10
```

```
Switch(config-if)# exit
```

! Assign FastEthernet0/2 to VLAN 20

```
Switch(config)# interface FastEthernet0/2
```

```
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
```

Explanation: The 'switchport mode access' command defines the port operational mode to carry untagged traffic for a single VLAN. Assigning Fa0/3 to VLAN 10 binds PC0 (Switch0) and PC2 (Switch1) into Logical Broadcast Domain 10. Similarly, assigning Fa0/2 to VLAN 20 binds PC1 (Switch0) and PC3 (Switch1) into Logical Broadcast Domain 20.

Step 4: Inter-Switch Trunk Link Configuration (Fa0/1)

To carry traffic for both VLAN 10 and VLAN 20 across the single physical crossover link between Switch0 and Switch1, interface FastEthernet0/1 was configured as an 802.1Q trunk port:

```
--- Trunk Configuration on Interface Fa0/1 ---
Switch(config)# interface FastEthernet0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
Switch# write memory
```

Explanation: The 'switchport mode trunk' command puts the interface into permanent trunking mode. The trunk link inserts an IEEE 802.1Q tag into the frame header when transmitting data across Switch0 and Switch1, allowing the receiving switch to identify which VLAN the frame belongs to.

Step 5: CLI Verification & Output Analysis

To confirm the VLAN database and trunk operational status, verification commands 'show vlan brief' and 'show interface trunk' were executed on the switches:

```
%SYS-5-CONFIG_I: Configured from console by console
Switch# show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                           Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/1, Gig0/2
10   VLAN0010                active    Fa0/3
20   VLAN0020                active    Fa0/2
1002 fddi-default            active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active
Switch#
```

Copy

Paste

```
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                           Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/1, Gig0/2
10   VLAN0010                active    Fa0/3
20   VLAN0020                active    Fa0/2
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch#
```

Copy Paste

```
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20

Switch#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch FastEthernet0/1 (10).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch FastEthernet0/1 (10).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch FastEthernet0/1 (10).

Switch#show vlan brief
```

Copy Paste

```
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface trunk
^
% Invalid input detected at '^' marker.

Switch#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
```

Copy Paste

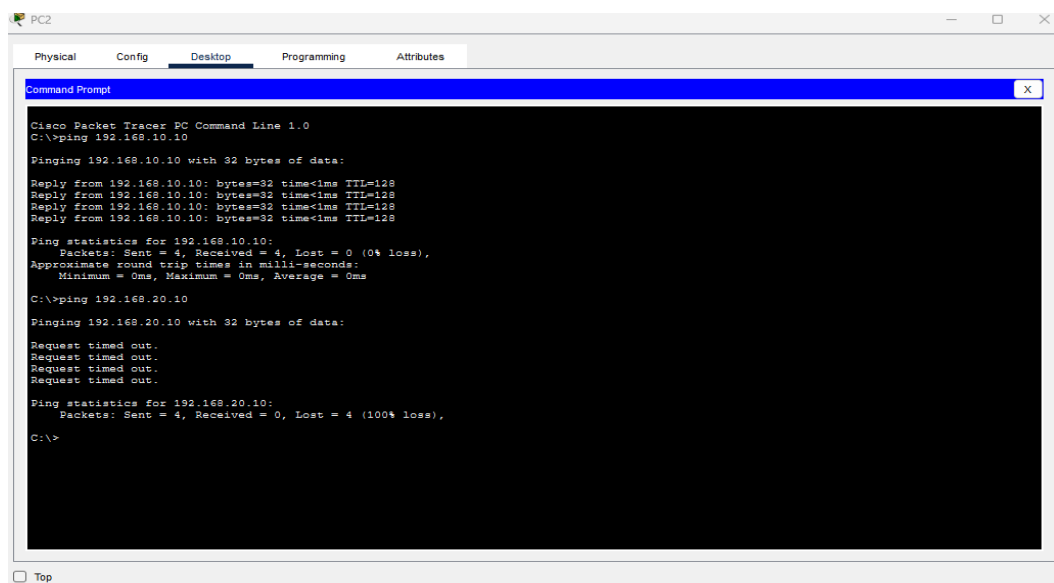
Analysis of CLI outputs:

1. 'show vlan brief' verifies that FastEthernet0/3 is active under VLAN 10, FastEthernet0/2 is active under VLAN 20, and FastEthernet0/1 is removed from default access VLAN 1 (because it operates as a trunk).
2. 'show interface trunk' confirms port Fa0/1 mode is 'on', encapsulation is '802.1q', status is 'trunking', and allowed active VLANs include 1, 10, and 20.

Connectivity Testing & Results

To test network functionality and inter-VLAN isolation, ICMP ping tests were executed from PC2 (192.168.10.20 - VLAN 10) in the Command Prompt:

- **Test 1: Ping to PC0 (192.168.10.10 - Same VLAN 10 across trunk)**
Result: SUCCESS. 4 packets sent, 4 packets received, 0% packet loss (RTT < 1ms). This proves that 802.1Q trunking successfully transports VLAN 10 traffic across switches.
- **Test 2: Ping to PC1 (192.168.20.10 - Different VLAN 20 across trunk)**
Result: FAILED / TIMED OUT. 4 packets sent, 0 received, 100% packet loss. This confirms complete Layer 2 broadcast domain isolation between VLAN 10 and VLAN 20 in the absence of a Layer 3 router.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

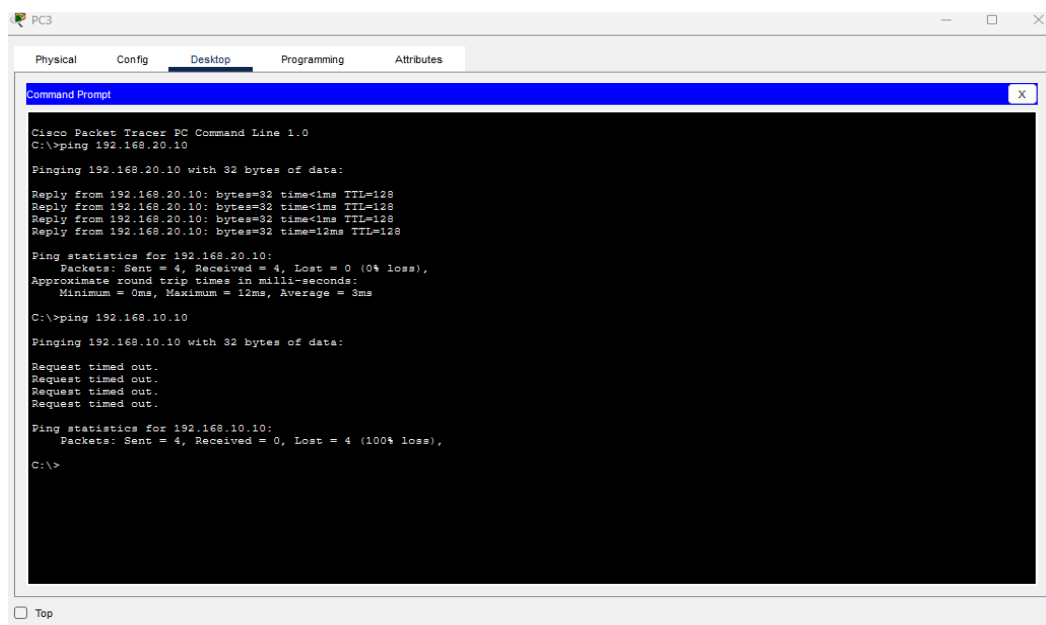
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=128
Reply from 192.168.20.10: bytes=32 time<1ms TTL=128
Reply from 192.168.20.10: bytes=32 time<1ms TTL=128
Reply from 192.168.20.10: bytes=32 time=12ms TTL=128

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Conclusion:

This experiment successfully demonstrated the implementation and operation of Virtual Local Area Networks (VLANs) and IEEE 802.1Q trunking across a multi-switch Cisco topology. The lab verified two key principles of enterprise networking:

- **Intra-VLAN Multi-Switch Communication:** By configuring interface Fa0/1 as an 802.1Q trunk, devices belonging to the same VLAN (such as PC0 and PC2 on VLAN 10) can communicate seamlessly across separate physical switches over a single physical cable link.
- **Inter-VLAN Security & Isolation:** Devices residing on different VLANs (such as VLAN 10 and VLAN 20) are logically isolated at Layer 2. Unscheduled inter-VLAN pings fail entirely, preventing unwanted cross-department broadcast traffic and reinforcing security without requiring additional physical hardware.